5. **(a)** Define *electric field strength*.

.....

.....

[1]

.....

(b) Electrons from a filament source enter a region between the parallel plates after being accelerated by an electric field. Fig. 5 below shows the electrons travelling horizontally at a speed of 2.50×10^7 m s^{-1} entering the pair of parallel plates.

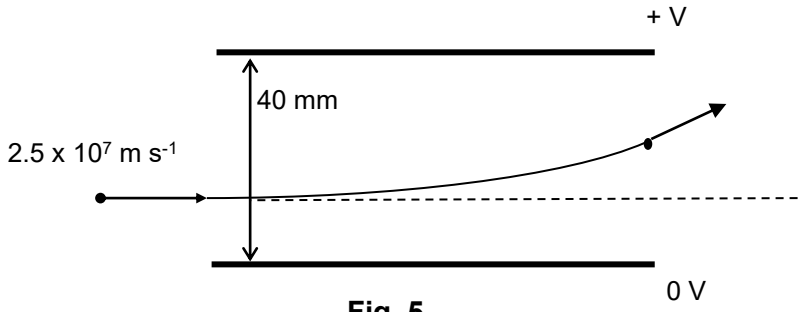

Fig. 5

(i) The electrons deviate by 30° on leaving the parallel plate of 80.0 mm long. The separation between the plates is 40.0 mm.

Calculate the time taken for the electrons to travel through the plates.

time = ... s [1]

(ii) Calculate the vertical component of the velocity when the electrons exit the parallel plates.

vertical component of velocity = ... m s^{-1} [2]

(iii) Hence calculate the acceleration of the electrons.

acceleration = ... m s^{-2} [1]

(iv) Calculate the potential difference *V* between the two plates.

V = ... V [2]

(v) If the plate separation is reduced, suggest how the answer to **(b)(ii)** is affected.

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..... [1]